

Table 4. Performance by Model Provider

Provider	Models	Pass Rate	Tests
x-ai	7	79.3%	140
sao10k	5	57.0%	100
anthropic	11	56.8%	220
cognitivecomputations	3	56.7%	60
cohere	4	56.2%	80
perplexity	3	55.0%	60
thedrummer	5	54.0%	100
google	21	52.1%	420
openai	36	48.3%	720
qwen	32	48.1%	640

of 331) of nominally "available" models had non-functional endpoints or access restrictions, underscoring the methodological importance of endpoint verification in preventing confounded results. These findings have important implications for both model development and deployment strategies.

Key Performance Patterns

Our primary hypothesis—that instruction-following capabilities would vary significantly across contemporary language models—is strongly supported by our results, which show performance ranging from 0% to 100% across the 256 tested models. Analysis of the evaluation results reveals several consistent patterns across the model landscape. First, we observe that model size does not consistently correlate with instruction-following capability. While larger models generally demonstrate superior performance on knowledge-intensive tasks, our results show that some smaller, specialized models outperform their larger counterparts on specific instruction-following tests. This confirms our secondary hypothesis that instruction adherence is not merely a function of parameter count but rather reflects specific training methodologies and optimization objectives.

Our hypothesis that multi-step sequential operations would prove challenging is validated by the results from Test 1 (Multi-step String Manipulation), which achieved only a 7.8% pass rate. Sequential instruction tasks proved particularly challenging for many models, with common failures occurring at transition points between steps. Models frequently executed individual steps correctly but failed to maintain coherence across the entire sequence. This pattern suggests that maintaining state across multi-step instructions remains a fundamental challenge, possibly reflecting limitations in how models process and maintain context through sequential operations.

Contrary to our expectation that simple constraint compliance would be universally manageable, content constraint tasks, exemplified by our letter-avoidance test (Test 2), revealed interesting dichotomies in model behavior. While Test 2 achieved a high 96.1% pass rate for exact output compliance, more complex constraints showed binary outcomes: models either succeeded completely or failed catastrophically, with few intermediate cases. Successful models appeared to implement robust checking mechanisms, while failing models showed no awareness of the constraint violation. This binary outcome distribution suggests that constraint awareness is either effectively incorporated during training or entirely absent, with little middle ground.

Format Compliance and Structured Output

Our hypothesis that structured output generation would reveal significant performance variations is confirmed by the format conversion category achieving only 57.8% average success. Our structured output tests, particularly the JSON formatting requirement (Test 3 with 14.8% pass rate), exposed significant variations in models' ability to produce machine-readable outputs. While most models could generate syntactically valid JSON, many failed to adhere to specific formatting requirements such as the absence of spaces or precise quote usage. This finding has direct implications for system integration, where precise format compliance is essential for downstream processing.

Supporting our hypothesis that training background influences instruction-following capabilities, models trained specifically for code generation showed superior performance on structured output tasks, even when the task itself was not explicitly code-related. This cross-domain transfer suggests that exposure to formal language structures during training enhances general instruction-following capabilities, particularly for tasks requiring precise syntactic compliance.

Model Family Characteristics

Our hypothesis that different model providers would show distinct instruction-following patterns is strongly supported, with provider performance ranging from 33.3% (Nvidia) to 79.3% (x-ai) among providers with at least 3 models. Our comprehensive evaluation reveals distinct characteristics among different model families. Anthropic's Claude models consistently demonstrated strong performance (56.8% average) across diverse instruction types, suggesting effective incorporation of instruction-following objectives in their training process. OpenAI's GPT series showed moderate performance (48.3% average) on standard tasks but occasionally struggled with highly specific formatting requirements. Open-source models exhibited greater variability, with some achieving comparable performance to proprietary models while others showed significant gaps in instruction adherence.

Confirming our hypothesis that instruction-tuning would improve performance, the evaluation revealed that instruction-tuned variants consistently outperformed their base model counterparts, validating the importance of specialized fine-tuning for instruction-following tasks. However, the degree of improvement varied significantly across model families, suggesting differences in instruction-tuning methodologies and datasets.

Implications for Model Development

Our hypothesis that specific training approaches would impact instruction-following capabilities is validated by several key findings. The binary nature of success on constraint-based tasks indicates that instruction-following capabilities benefit from explicit training on constraint satisfaction rather than emerging purely from scale. The consistent failures at task transition points in sequential instructions (as seen in our 7.8% pass rate for Test 1) suggest that training data should emphasize multi-step procedures with explicit state maintenance requirements.

Supporting our hypothesis about cross-domain transfer, the strong performance of code-trained models on structured output tasks suggests that incorporating formal language data during pretraining may enhance general instruction-following capabilities. This finding supports a more diverse pretraining strategy that includes both natural language and formal language structures.

Practical Applications and Deployment Considerations

For practitioners deploying LLMs in production environments, our results highlight the importance of thorough instruction-following evaluation before deployment. The significant variability in performance, even among models with similar benchmark scores on other tasks, suggests that standard evaluations may not adequately capture instruction adherence capabilities.

The format compliance results have particular relevance for system integration scenarios. Applications requiring structured outputs should implement robust validation layers rather than assuming model compliance, even when using high-performing models. The binary nature of constraint satisfaction performance suggests that fallback mechanisms are essential for critical applications.

Limitations of the Current Study

While our evaluation framework provides valuable insights, several limitations should be acknowledged. The selection of twenty diagnostic prompts, while carefully designed, cannot capture the full spectrum of instruction-following scenarios encountered in real-world applications. Our focus on verifiable, objective criteria necessarily excludes more nuanced forms of instruction following that may be equally important but harder to assess automatically.

The evaluation’s reliance on the OpenRouter API, while providing consistent access to multiple models, may introduce platform-specific biases or limitations. Some models may perform differently when accessed through their native APIs or with different parameter settings. Additionally, our timeout constraints may have disadvantaged some models that require longer processing times for complex instructions.

Limitations

- **Language Coverage:** English-only
- **Limited Scope:** Focuses on verifiable instructions only, may not capture all aspects of instruction-following
- **Single-Turn Focus:** Does not evaluate multi-turn conversations

Future Directions

This work opens several avenues for future research. Expanding the diagnostic prompt set to include additional instruction categories could provide more comprehensive coverage while maintaining the framework’s efficiency. Investigating the relationship between instruction-following performance and other model capabilities could reveal underlying connections and trade-offs in model design.

Developing adaptive evaluation frameworks that adjust prompt complexity based on model performance could provide more nuanced assessments of instruction-following capabilities. Such frameworks could identify the precise boundaries of model capabilities rather than binary pass/fail outcomes.

Further research into the training methodologies that promote strong instruction-following would be valuable. Understanding whether specific data characteristics, training objectives, or architectural choices contribute to superior instruction adherence could inform future model development.